Course Introduction

AWS Academy Cloud Foundations

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Module overview

Topics

- Course objectives and overview
- AWS certification exam information
- AWS Documentation

Activities

- AWS Documentation scavenger hunt

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Module objectives

After completing this module, you should be able to:

- Recognize the purpose of the AWS Academy Cloud Foundations course
- Recognize the course structure
- Recognize the AWS certification process
- Navigate the AWS Documentation website

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Section 1: Course objectives and overview

Course Introduction

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Course prerequisites

- General Required Knowledge
  - IT technical knowledge
  - IT business knowledge
- Preferred Knowledge
  - Familiarity with cloud computing concepts
  - Working knowledge of distributed systems
  - Familiarity with general networking concepts
  - Working knowledge of multi-tier architectures

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Course objectives

After completing this course, you should be able to:

- Define the AWS Cloud.
- Explain the AWS pricing philosophy.
- Identify the global infrastructure components of AWS.
- Describe security and compliance measures of the AWS Cloud including AWS Identity and Access Management (IAM).
- Create an AWS Virtual Private Cloud (Amazon VPC).
- Demonstrate when to use Amazon Elastic Compute Cloud (EC2), AWS Lambda and AWS Elastic Beanstalk.
- Differentiate between Amazon S3, Amazon EBS, Amazon EFS and Amazon S3 Glacier.
- Demonstrate when to use AWS Database services including Amazon Relational Database Service (RDS), Amazon DynamoDB, Amazon Redshift, and Amazon Aurora.
- Explain AWS Cloud architectural principles.
- Explore key concepts related to Elastic Load Balancing (ELB), Amazon CloudWatch, and Auto Scaling.

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Course outline

- Module 1: Cloud Concepts Overview
- Module 2: Cloud Economics and Billing
- Module 3: AWS Global Infrastructure Overview
- Module 4: AWS Cloud Security
- Module 5: Networking and Content Delivery
- Module 6: Compute
- Module 7: Storage
- Module 8: Databases
- Module 9: Cloud Architecture
- Module 10: Automatic Scaling and Monitoring

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Module 1: Cloud Concepts Overview

Module sections:

- Introduction to cloud computing
- Advantages of cloud computing
- Introduction to Amazon Web Services (AWS)
- Moving to the AWS Cloud – The AWS Cloud Adoption Framework (AWS CAF)

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Module 2: Cloud Economics and Billing

Module sections:

- Fundamentals of pricing
- Total Cost of Ownership
- AWS Organizations
- AWS Billing and Cost Management
- Technical support

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Module 3: AWS Global Infrastructure Overview

Module sections:

- AWS Global Infrastructure
- AWS services and service category overview

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Module 4: AWS Cloud Security

Module sections:

- AWS shared responsibility model
- AWS Identity and Access Management (IAM)
- Securing a new AWS account
- Securing accounts
- Securing data on AWS
- Working to ensure compliance

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Module 5: Networking and Content Delivery

Module sections:

- Networking basics
- Amazon VPC
- VPC networking
- VPC security
- Amazon Route 53
- Amazon CloudFront

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Module 6: Compute

Module sections:

- Compute services overview
- Amazon EC2
- Amazon EC2 cost optimization
- Container services
- Introduction to AWS Lambda
- Introduction to AWS Elastic Beanstalk

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Module 7: Storage

Module sections:

- Amazon Elastic Block Store (Amazon EBS)
- Amazon Simple Storage Service (Amazon S3)
- Amazon Elastic File System (Amazon EFS)
- Amazon Simple Storage Service Glacier

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Module 8: Databases

Module sections:

- Amazon Relational Database Service (Amazon RDS)
- Amazon DynamoDB
- Amazon Redshift
- Amazon Aurora

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Module 9: Cloud Architecture

Module sections:

- AWS Well-Architected Framework
- Reliability and availability
- AWS Trusted Advisor

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Module 10: Automatic Scaling and Monitoring

Module sections:

- Elastic Load Balancing
- Amazon CloudWatch
- Amazon EC2 Auto Scaling

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Section 2: AWS certification exam information

Course Introduction

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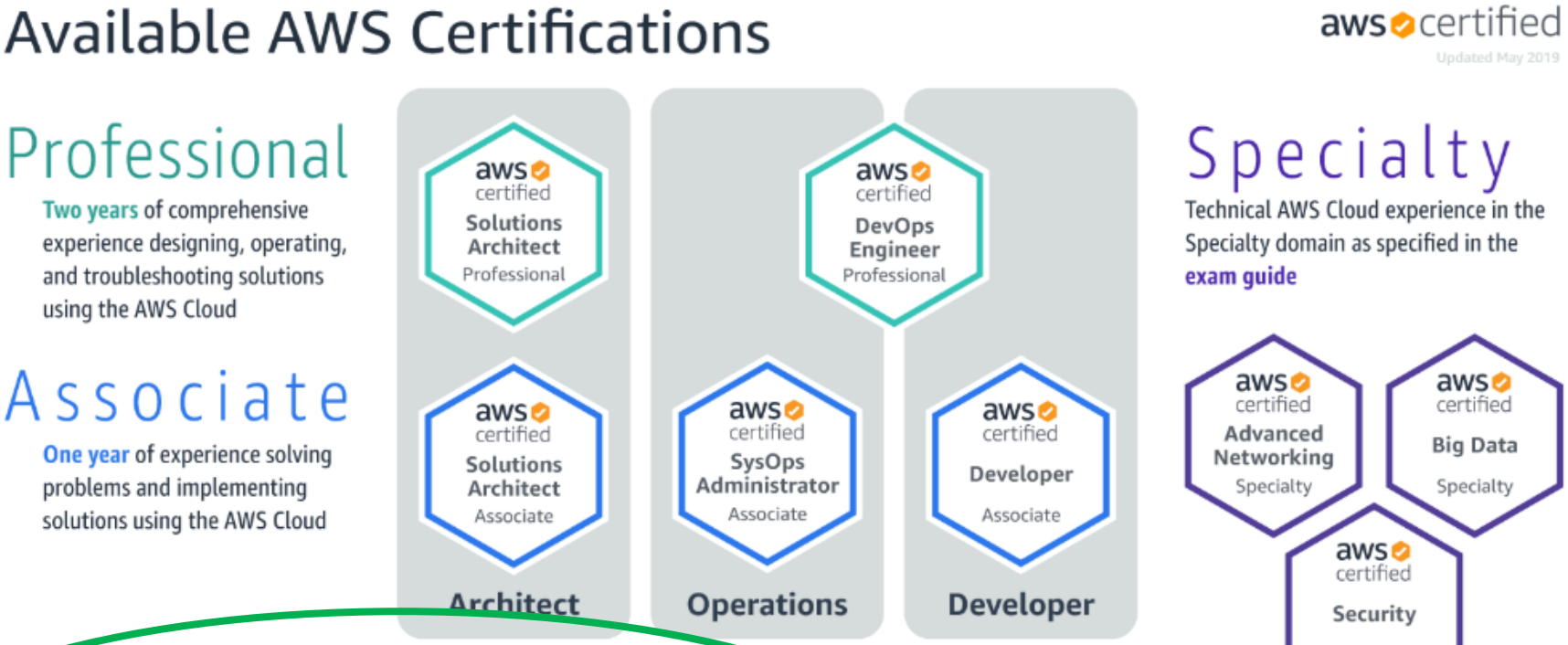
AWS certification exams



*This course helps prepare you for the AWS **Cloud Practitioner** certification exam*

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AWS Certified Cloud Practitioner exam

- Details about the exam—including how to register for it—are at https://aws.amazon.com/certification/certified-cloud-practitioner/
  - Download and carefully read the AWS Certified Cloud Practitioner Exam Guide at https://d1.awsstatic.com/training-and-certification/Docs - Cloud Practitioner/AWS_Certified_Cloud_Practitioner-Exam_Guide_EN_v1.6.pdf
  - Download the sample exam questions at https://d1.awsstatic.com/training-and-certification/Docs - Cloud Practitioner/AWS Certified Cloud Practioner_Sample Questions_v1.1_FINAL.PDF

- See the recommended path to attain the certification at https://aws.amazon.com/training/path-cloudpractitioner/
  - AWS Academy Cloud Foundations covers much of the same material found in the Cloud Practitioner Essentials course, but in greater depth.
  - There is additional free digital training available at https://www.aws.training/.

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Section 3: AWS Documentation

Course Introduction

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AWS Documentation

- Find user guides, developer guides, API references, tutorials, and more at https://docs.aws.amazon.com/

- ***Whitepapers*** are also available at https://aws.amazon.com/whitepapers/, including these which are recommended reading for the AWS Cloud Practitioner exam:
  - Overview of Amazon Web Services: https://d0.awsstatic.com/whitepapers/aws-overview.pdf
  - Architecting for the Cloud: AWS Best Practices: https://d1.awsstatic.com/whitepapers/AWS_Cloud_Best_Practices.pdf
  - How AWS Pricing Works: https://d0.awsstatic.com/whitepapers/aws_pricing_overview.pdf
  - The Total Cost of (Non) Ownership of Web Applications in the Cloud: https://media.amazonwebservices.com/AWS_TCO_Web_Applications.pdf

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